



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

EndSemester

Semester: III

Programme: B. Tech Information Technology

Course: Data Structure

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32301	Course Outcome
CO1	To gain knowledge about basic concepts of data structures.
CO2	To acquire knowledge of stacks and queues with their applications.
CO3	To aware about the concepts of trees with their applications.
CO4	To learn and design the algorithm for graphs with their applications.
CO5	To implement basic operations on linked list.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Non-Linear Data Structure.	CO1	2	2
b.	Define Circular Linked List.	CO2	2	2
c.	Define Stack Underflow.	CO3	2	2
d.	What is Adjacency Matrix?	CO4	3	2
e.	Define AVL Tree.	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain Bubble Sort with suitable example.	CO1	3	5
b.	Explain Binary Search algorithm with example.	CO1	3	5
c.	Compare Bubble Sort, Quick Sort and Merge Sort.	CO1	4	5
Que. 3	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain types of Linked List with neat diagrams.	CO2	4	5
b.	Explain insertion and deletion in Doubly Linked List.	CO2	3	5
c.	Explain advantages and disadvantages of Linked List.	CO2	2	5
Que. 4	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain representation of Stack using Array.	CO3	2	5
b.	Explain Queue operations Insert and Delete.	CO3	4	5
c.	Explain expression evaluation using Stack.	CO3	3	5
Que 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain graph representation techniques.	CO4	2	5
b.	Compare DFS and BFS traversals.	CO4	3	5
c.	Explain Kruskal's Algorithm with example.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Binary Tree Traversals with example.	CO5	2	5
b.	Explain insertion and deletion in BST.	CO5	3	5
c.	Explain Threaded Binary Tree Traversals.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create